

August 22, 2026

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2023305556

BS Data Science DS3A

DS413 Elective 4 (Deep Learning)

Homework 1.1

Deep Learning Technique: *Transformer Network.*

First Appeared: *June 2017.*

Purpose:

The Transformer architecture was created to replace Recurrent Neural Networks (RNNs) and Long Short-Term Memory networks (LSTMs) for sequence-to-sequence tasks. They process sequential data simultaneously rather than sequentially, enabling much faster computational throughput and better capture of long-range dependencies across context windows.

Key Innovations:

1. ***Self-Attention Mechanisms:*** Instead of processing sequence items step-by-step or word-by-word, self-attention evaluates the relative relationships between every element in a sequence at once.
2. ***Massive Parallelization:*** Without any recurrent hidden state loops, training can be distributed across thousands of GPU/TPU cores simultaneously.
3. ***Positional Encodings:*** Positional vector signatures are injected into token embeddings, allowing the model to understand token order without needing sequential recurrence.

How It Works:

Input Tokens -> Embedding + Positional Encoding -> Multi-Head Self-Attention -> Feed Forward NN -> Output.

1. ***Embedding & Positional Encoding:*** The input tokens, such as words or sub-words, are converted into dense vector representations. Sinusoidal or learned positional encodings are added to preserve sequential context.
2. ***Linear Projections:*** Each input vector is transformed into three dynamic matrices via linear projections:
 - **Query (Q):** Represents what a token is searching for.
 - **Key (K):** Represents what a token offers.
 - **Value (V):** Represents the actual contextual information.

3. **Scaled Dot-Product Attention:** The interaction matrix is computed using the attention equation:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

4. **Multi-Head Attention:** Multiple attention mechanisms operate in parallel across different feature subspaces, allowing the model to process syntax, semantics, and context simultaneously.
5. **Feed-Forward and Normalization:** The outputs pass through residual connections, layer normalization, and position-wise feed-forward networks to construct final context representations.

Real-World Application:

Large Language Models & Generative AI: Modern models, such as GPT-4, Claude, LLaMA, and Gemini, use decoder-based Transformer architectures for text generation, automated code completion, multi-turn reasoning, and language translation.

References:

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2. Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., & Houlsby, N. (2020). An image is worth 16x16 words: Transformers for image recognition at scale. *International Conference on Learning Representations (ICLR)*. <https://arxiv.org/abs/2010.11929>